

Lecture 25 Divergence Theorem Cont'd

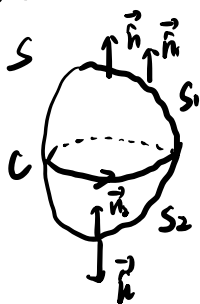
Recap: Divergence Theorem

$$\underbrace{\oint_S \vec{F} \cdot \vec{n} \, ds}_{\text{surface integral}} = \underbrace{\iiint_D \nabla \cdot \vec{F} \, dV}_{\text{triple integral}}$$

Stokes' Theorem $\underbrace{\oint_C \vec{F} \cdot d\vec{r}}_{\text{line integral}} = \underbrace{\iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds}_{\text{surface integral}}$

$$\iiint_D \nabla \cdot (\nabla \times \vec{F}) \, dV = 0$$

$$\oint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds = 0 \quad \leftarrow \quad \oint_S (\nabla \times \vec{F}) \cdot \vec{n}_1 \, ds - \oint_S (\nabla \times \vec{F}) \cdot \vec{n}_2 \, ds = 0$$



Interpretation of divergence operator.

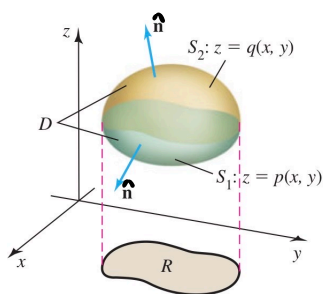
By taking a tiny solid region D with boundary surface S

$$\oint_S \vec{F} \cdot \vec{n} \, ds = \iiint_D \nabla \cdot \vec{F} \, dV \approx (\nabla \cdot \vec{F}) \iiint_D dV = (\nabla \cdot \vec{F}) \times \text{volume of } D$$



$$\nabla \cdot \vec{F} \approx \frac{\oint_S \vec{F} \cdot \vec{n} \, ds}{\text{volume of } D} \quad \text{measures the flux density near a point}$$

Proof of the Divergence Theorem.



$$\vec{F} = \langle F_x, F_y, F_z \rangle$$

$$\oint_S \vec{F} \cdot \vec{n} \, ds = \oint_S (F_x \vec{i} + F_y \vec{j} + F_z \vec{k}) \cdot \vec{n} \, ds$$

$$= \oint_S F_x \vec{i} \cdot \vec{n} \, ds + \oint_S F_y \vec{j} \cdot \vec{n} \, ds + \oint_S F_z \vec{k} \cdot \vec{n} \, ds$$

$$\iiint_D \nabla \cdot \vec{F} \, dV = \iiint_D \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \right) dV$$

To show $\oint_S F_z \vec{k} \cdot \vec{n} \, ds = \iiint_V \frac{\partial F_z}{\partial z} \, dV$ $\vec{F} = \langle 0, 0, F_z \rangle$

$$\iiint_V \frac{\partial F_z}{\partial z} \, dV = \iint_R \int_{p(x,y)}^{q(x,y)} \frac{\partial F_z}{\partial z} \, dz \, dx \, dy$$

$$= \iint_R (F_z(x, y, q(x, y)) - F_z(x, y, p(x, y))) \, dx \, dy$$

By the Fundamental Theorem of Calculus.

The outward normal to S_2 is $\langle -\frac{\partial q}{\partial x}, -\frac{\partial q}{\partial y}, 1 \rangle$

The outward normal to S_1 is $\langle \frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, -1 \rangle$

$$\oint_S F_z \vec{k} \cdot \vec{n} \, ds = \iint_{S_2} F_z \vec{k} \cdot \vec{n} \, ds + \iint_{S_1} F_z \vec{k} \cdot \vec{n} \, ds$$

$$= \iint_R F_z(x, y, q(x, y)) \vec{k} \cdot \langle -\frac{\partial q}{\partial x}, -\frac{\partial q}{\partial y}, 1 \rangle \, dx \, dy$$

$$+ \iint_R F_z(x, y, p(x, y)) \vec{k} \cdot \langle \frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, -1 \rangle \, dx \, dy.$$

$$= \iint_R F_z(x, y, q(x, y)) \, dx \, dy - \iint_R F_z(x, y, p(x, y)) \, dx \, dy.$$

$$\left(\begin{aligned} \iint_S \vec{F} \cdot \vec{n} \, ds &= \pm \iint_R \vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) \, du \, dv. \\ &= \pm \iint_R \left(-F_x \frac{\partial z}{\partial x} - F_y \frac{\partial z}{\partial y} + F_z \right) \, dx \, dy. \end{aligned} \right) \quad \square$$

Limitations of the Divergence Theorem

$$\vec{F} = -\frac{\gamma M m}{(x^2 + y^2 + z^2)^{3/2}} \frac{x\vec{i} + y\vec{j} + z\vec{k}}{(x^2 + y^2 + z^2)^{3/2}}$$

$$(p, \theta, \phi): \quad \rho^2 = x^2 + y^2 + z^2$$

$$\vec{e}_\rho = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}} \text{ is the unit radial vector}$$

$$\vec{F} = -\frac{\gamma M m}{\rho^2} \vec{e}_\rho$$

S is the sphere with radius a centered at the origin

$$\rho = a, \quad \vec{n} = \vec{e}_\rho$$

$$\vec{F} \cdot \vec{n} = -\frac{Gmm}{\rho^2} \vec{e}_\rho \cdot \vec{e}_\rho = -\frac{Gmm}{\rho^2}$$

$$\oint_S \vec{F} \cdot \vec{n} ds = \oint_S -\frac{Gmm}{\rho^2} ds = \oint_S -\frac{Gmm}{a^2} ds = -\frac{Gmm}{a^2} \oint_S ds$$

On the other hand, $\nabla \cdot \vec{F} = 0$

$$\iiint_{D_0} \nabla \cdot \vec{F} dv = 0$$

(except origin)

$$= -\frac{Gmm}{a^2} 4\pi a^2$$

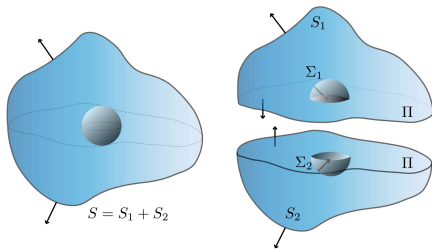
$$= -4\pi Gmm$$

$$\oint_S \vec{F} \cdot \vec{n} ds \neq \iiint_{D_0} \nabla \cdot \vec{F} dv.$$

Gauss's Law for gravity

$$\oint_S -\frac{Gmm}{\rho^2} \vec{e}_\rho \cdot \vec{n} ds = -4\pi Gmm \text{ for any closed surface } S \text{ enclosing the origin}$$

$$\oint_S \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds = 4\pi$$



S_1, Π, Σ_1 is a closed surface; D_1 is the solid enclosed

$$(\iint_{S_1} + \iint_{\Pi} + \iint_{\Sigma_1}) \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds = \iiint_{D_1} \underbrace{\nabla \cdot \left(\frac{\vec{e}_\rho}{\rho^2} \right)}_0 dv = 0$$

$$\iint_{S_1} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n}_{up} ds + \iint_{\Pi} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n}_{down} ds + \iint_{\Sigma_1} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n}_{down} ds = 0 \quad (1)$$

$$\iint_{S_2} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n}_{down} ds + \iint_{\Pi} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n}_{up} ds + \iint_{\Sigma_2} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n}_{up} ds = 0 \quad (2)$$

$$(1) + (2): \oint_S \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds - \oint_S \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds = 0$$

$$\oint_S \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds = \oint_S \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds = 4\pi$$

$$\text{Gauss's Law: } \oint_S -\frac{1}{4\pi\epsilon_0} \frac{\vec{e}_\rho}{\rho^2} \cdot \vec{n} ds = \begin{cases} -\frac{1}{4\pi\epsilon_0} \frac{q}{\epsilon_0} & \text{if } S \text{ encloses the origin} \\ 0 & \text{otherwise.} \end{cases}$$

Theorem (Divergence Theorem for Hollow Regions) Let D be a simply-connected region in $\mathbb{R}^3$ with n holes. Denote S to be its outer boundary and $T_1, T_2, \dots, T_n$ to be its inner boundaries. Suppose $\vec{F}$ is defined and C^1 in and near the region D , then we have

$$\iiint_D \nabla \cdot \vec{F} dV = \oint_S \vec{F} \cdot \vec{n} ds - \sum_{i=1}^n \oint_{T_i} \vec{F} \cdot \vec{n} ds.$$

Here $\vec{n}$ is the outward unit normal for each boundary surface.

A Final Perspective

The cumulative/integrated effect of the derivatives of a function throughout a region is determined by the values of the function on the boundary.

Fundamental Theorem of Calculus

$$\int_a^b f'(x) dx = f(b) - f(a)$$



Fundamental Theorem of Line Integrals

$$\int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$



$$\text{Stokes' Theorem } \iint_S (\nabla \times \vec{F}) \cdot \vec{n} ds = \oint_C \vec{F} \cdot d\vec{r}$$

$$\text{Divergence Theorem } \iiint_D \nabla \cdot \vec{F} dV = \oint_S \vec{F} \cdot \vec{n} ds$$